

Troubleshooting Cables and Physical Layer — Guided

Notes ANSWER KEY

Unit 5.2.1 | CompTIA Network+ N10-009 Obj. 5.2 | N10-008 5.2

For instructor use / distribute after students complete the notes.

Question 1

A split pair passes a basic continuity test because pin-to-pin connectivity is _____, but the two conductors are not properly _____ together, destroying interference cancellation. Only a _____ tester that measures crosstalk will catch it.

Answer: Correct; twisted (paired); certification.

Question 2

Attenuation is measured in _____ and increases with cable _____. The maximum recommended run length for Cat6 copper cable is _____ meters total (including patch cables at each end).

Answer: Decibels (dB); length (and temperature); 100 meters.

Question 3

NEXT (Near-End Crosstalk) is caused by signal coupling between pairs at the _____ end of the cable. The most common installation cause is untwisting more than _____ of wire at the connector termination.

Answer: Transmitting (near/source); half an inch (about 13mm).

Question 4

Data cables running parallel to power cables should maintain at least _____ of separation. Where they must cross, they should cross at _____ to minimize electromagnetic coupling. If separation is impossible, use _____ cabling.

Answer: 12 inches (30cm); 90 degrees; STP (shielded twisted pair).

Question 5

A PoE port that repeatedly powers a device on and off is likely due to insufficient _____ on the switch. A device spec'd for 802.3at (PoE+, _____ watts) will fail on a switch that only supports 802.3af (_____ watts).

Answer: Power budget; 30W; 15.4W.

Question 6

Fiber optic bend radius violations occur when cable is bent tighter than the minimum specification, causing _____ to escape the fiber core (macrobending). The tool that makes these leaks visible is a _____.

Answer: Light; visual fault locator (VFL).

Question 7

A switch port with a green link light but rising CRC error counts in `show interface` output indicates a _____ layer problem. Name three symptoms a user might report that actually trace back to CRC errors.

Answer: Physical (Layer 1); symptoms: slow file transfers, dropped VoIP calls, intermittent connection timeouts, video buffering, high retransmission rates.

Question 8 — Real World Application

REAL WORLD: A newly installed patch panel in a wiring closet works for most ports, but four ports on one patch strip have poor throughput and high error rates at gigabit speeds, while basic link works fine at 100 Mbps. The cable runs are all under 15 meters. What is the most likely cause and how do you confirm it?

Answer: Split pairs on those four terminations — the conductor pairs were not punched down in the correct T568A or T568B sequence, so the twist pairing is broken. Basic link negotiates because continuity is intact, but gigabit (which uses all four pairs) exposes the crosstalk. Confirm with a certification tester measuring NEXT on each affected port.